MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG5DSECSC304 – Design Principles of User Computer Interaction

Duration: 2 Hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer All Questions
Each question carries 2 marks

1. List the different input-output channels involved in human-computer interaction. (K, CO1)
2. Describe the Model Human Processor (MHP). (U, CO2)
3. State Norman's seven stages of action. (K, CO3)
4. Explain any three of Shneiderman's eight golden rules of interface design. (U, CO3)
5. Define ergonomics in the context of interaction design. (K, CO4)

[5x2 = 10]

Part B
Short Answer Questions
Answer any 6 Questions
Each question carries 5 marks

6. List the three stages of human memory. (K, CO1)
7. State any four psychological principles applied in interactive system design. (K, CO1)
8. Identify the various input devices used in modern computing systems. (K, CO1)
9. Name the key design considerations for developing a navigation application for visually impaired users. (K, CO1)
10. Differentiate between heuristic evaluation and usability testing. (An, CO2)
11. Illustrate a suitable dialog design notation to represent a login interface. (A, CO3)
12. Discuss the major security considerations in human-computer interaction. (U, CO4)
13. Illustrate a framework for implementing groupware in a collaborative application. (A, CO4)

[6x5 = 30]

Part B
Short Answer Questions
Answer any 2 Questions
Each question carries 15 marks

14. Examine the process of usability testing for a task management application, comparing heuristic evaluation with contextual evaluation techniques. (An, CO2)
15. Discuss face-to-face and text-based communication in HCI, and summarize their impact on analyzing user feedback in collaborative video conferencing tools. (U, CO3)
16. Illustrate, with a suitable design, how Augmented and Virtual Reality can be employed in developing a health monitoring wearable device, addressing associated user privacy concerns. (A, CO4)

[2x15 = 30]